

System Documentation

Campus Food Ordering and Management System

Version 3.0 | Tutorial Section TT5L | Group 1 | 16/06/2026

Name	Student No.	Actor
Hassan Abdelhaleem Hassan Basheir	251UC2506T	Student
Rashad Ali Qaid Sofan	251UC2505G	Vendor
Ahmad Aljammal	251UC2500R	Admin
Mohammed A. M. Alakklouk	251UC250CR	Staff

1 Project Management

The project digitizes campus dining operations by supporting ordering, scheduling, mock payment, real-time queue tracking, admin-created vendor/staff accounts, vendor management, admin reporting, and staff pickup confirmation.

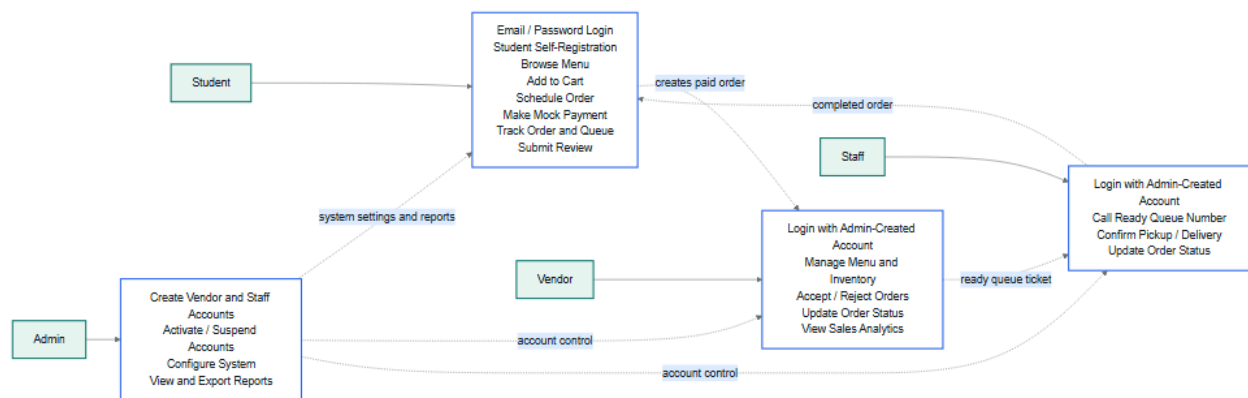
2 System Overview

Actors are Student, Vendor, Admin, and Staff. Login uses email/password only; the backend determines role from the database. Each dashboard maps to the responsibilities defined in Part I and Part II.

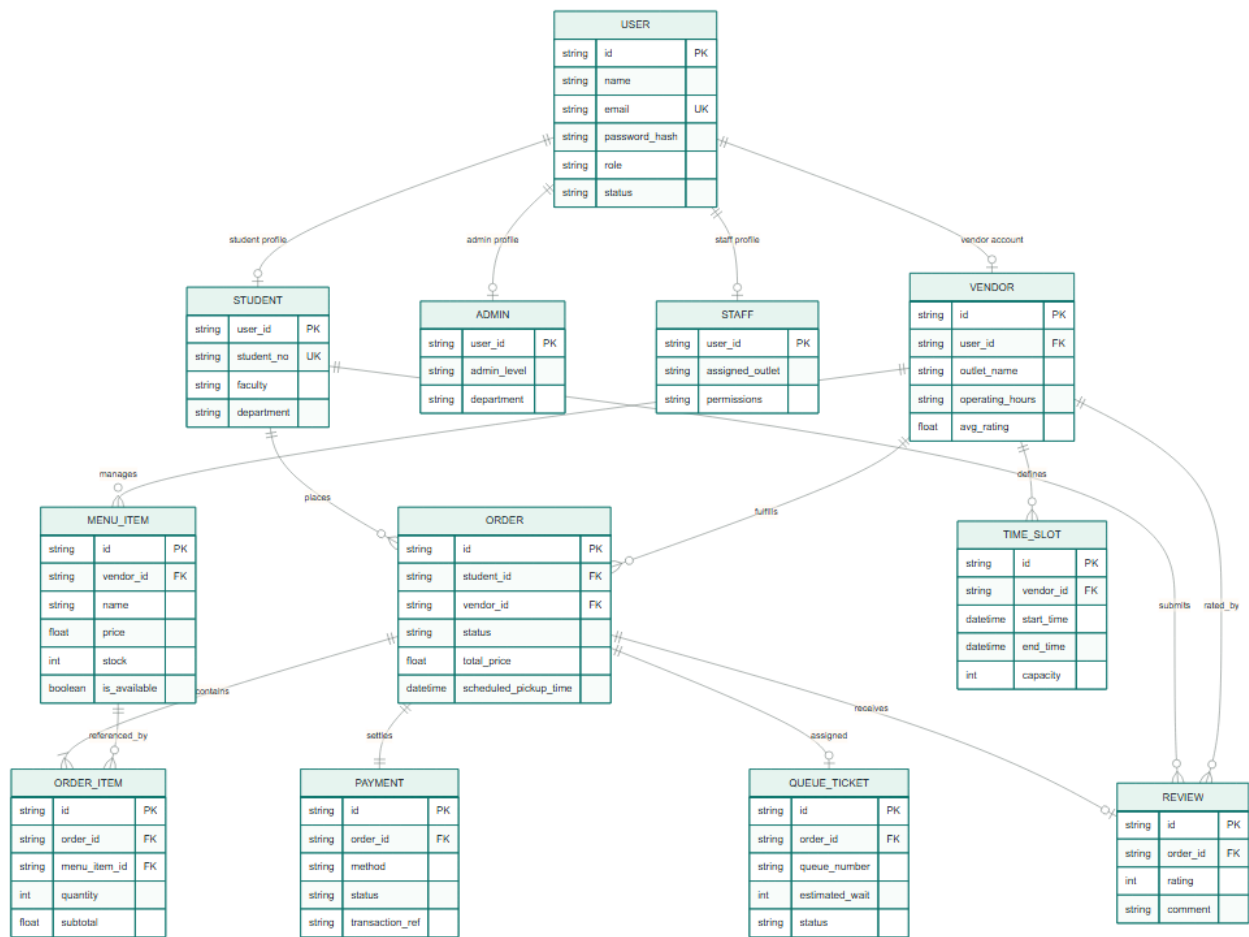
3 Requirements

The ERD and state model below summarize the implemented data and order lifecycle requirements.

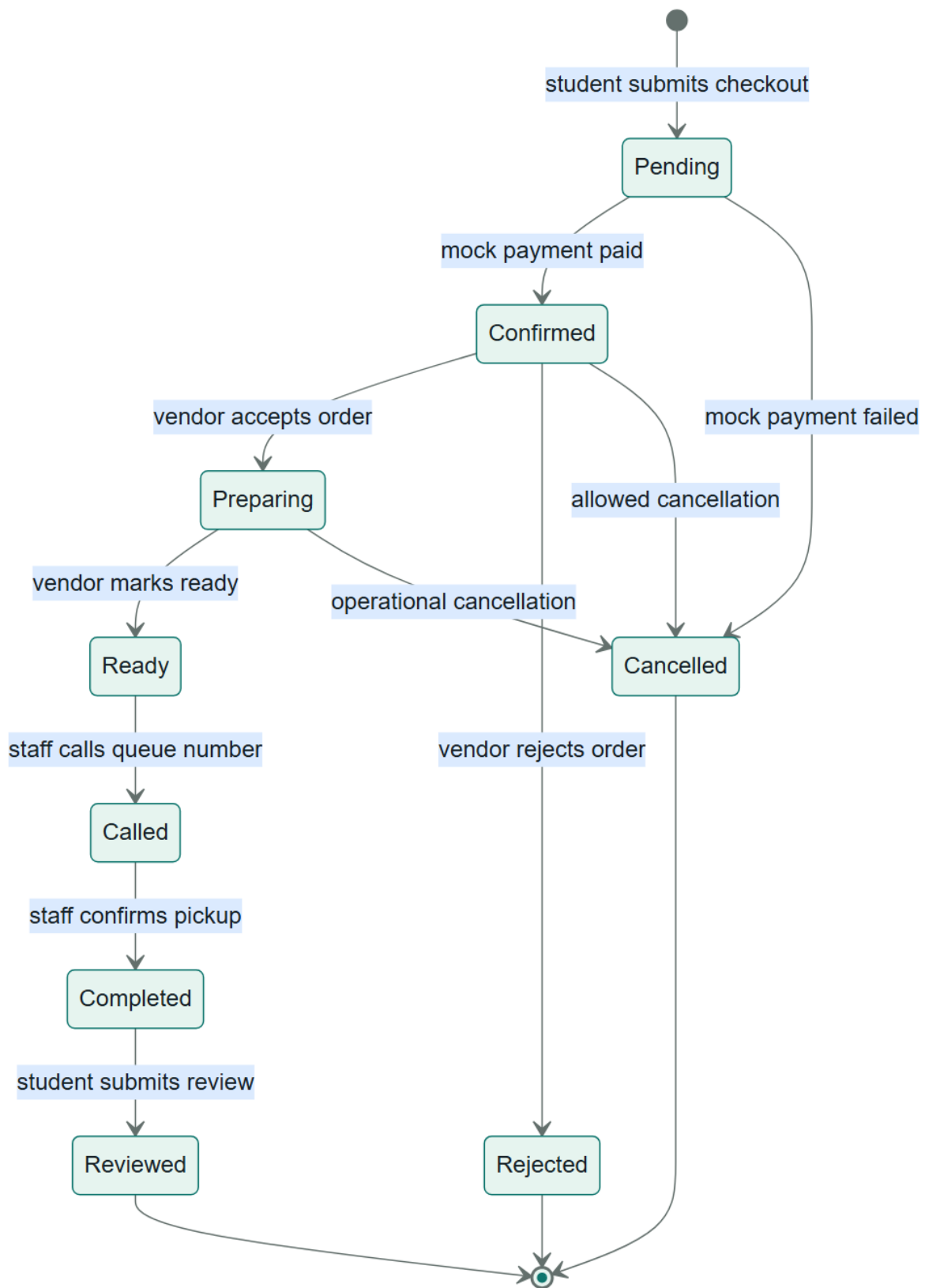
Use case diagram



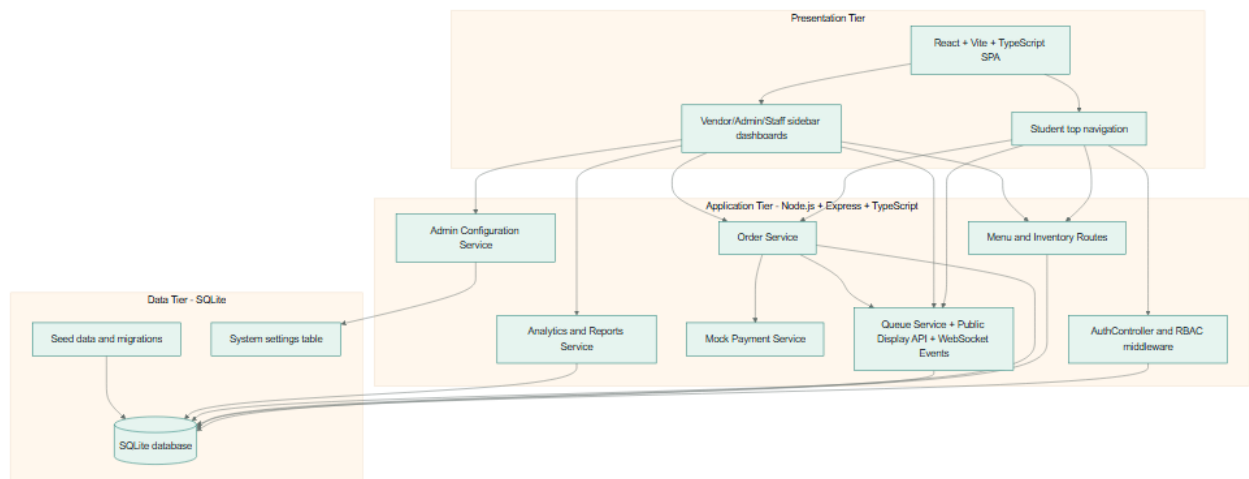
ERD/class diagram



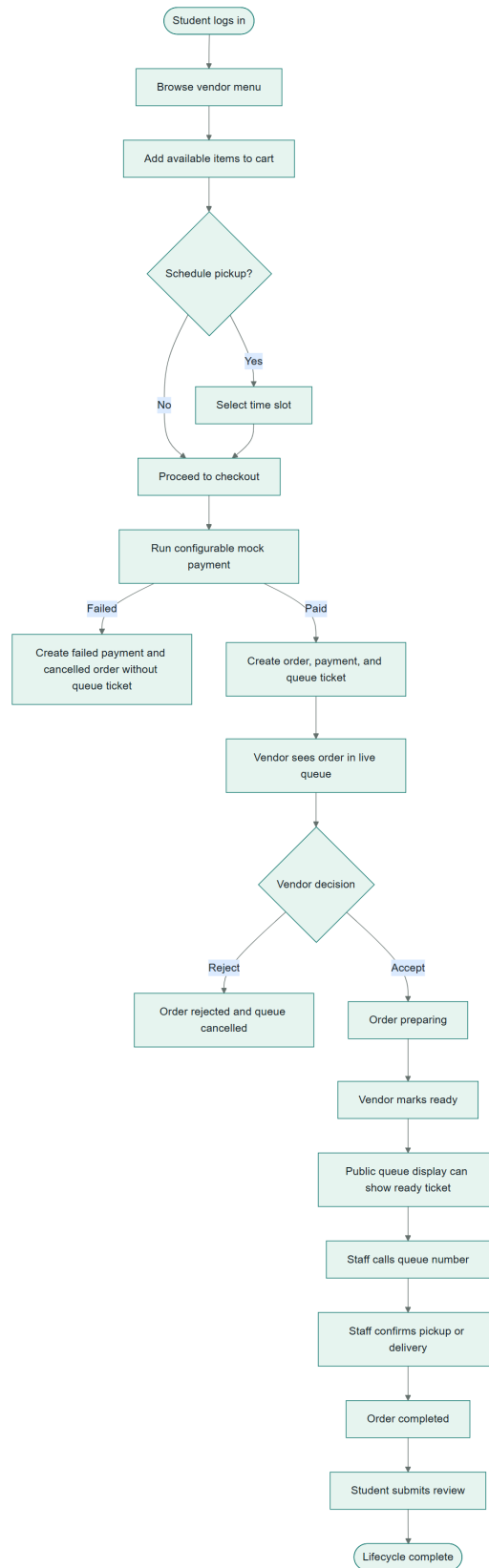
Order state diagram



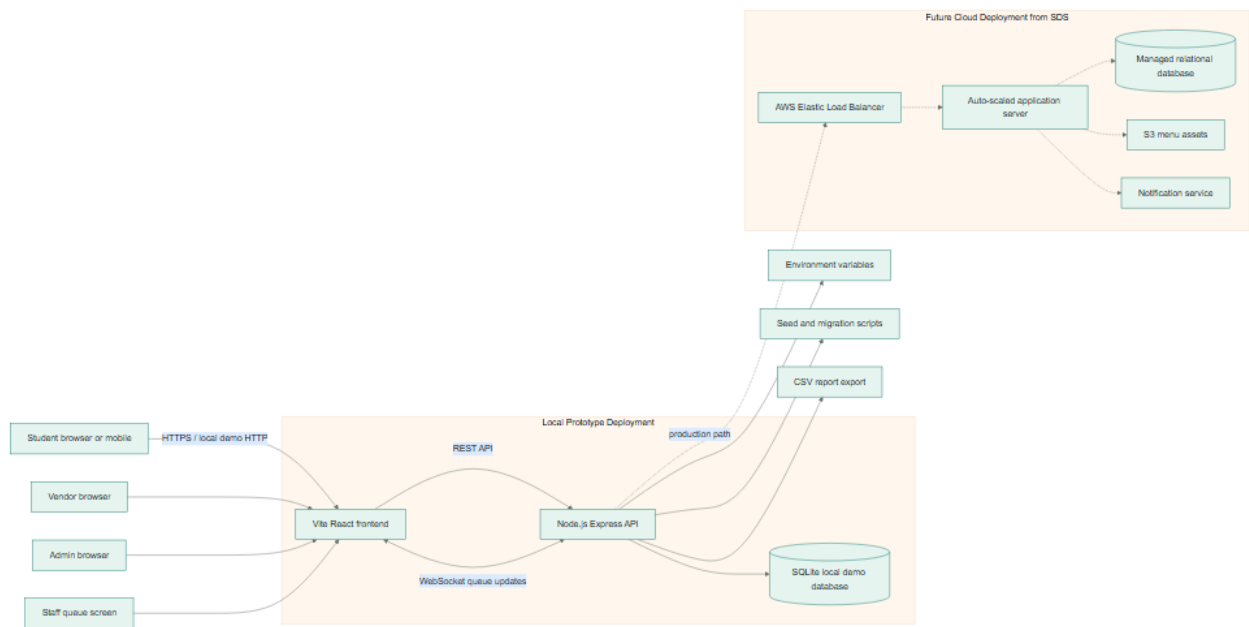
Software architecture



Order workflow



Deployment diagram



4 Design and Sample Screens

Email/password login, student registration, and public queue display



[Login](#) [Register](#)

Email

Password

[Enter Dashboard →](#)

Student menu

STUDENT ORDERING

Browse campus menus

Q Search meals, drinks, categories

LIVE KITCHEN QUEUE

Order ahead and pick up when
your queue number is called.

6

AVAILABLE ITEMS

Vendor

All vendors

Category

All categories

Burgers

ACTIVE

Crispy Chicken Burger

Crispy chicken fillet, campus slaw,
pickles, and house sauce.

⌚ 14 min ☆ 5.0 Campus Bites

RM 13.50

Add

Dessert

SUSPENDED

Sold Out Brownie Box

Chocolate brownie snack box for
afternoon breaks.

⌚ 3 min ☆ 5.0 Campus Bites

RM 5.50

Add

Drinks

ACTIVE

Iced Matcha Latte

Cold matcha latte with oat milk
option and light vanilla syrup.

⌚ 4 min ☆ 5.0 Campus Bites

RM 6.80

Add

Grab & Go

ACTIVE

Campus Veggie Wrap

Wholegrain wrap with hummus,
roasted vegetables, greens, and
yogurt sauce.

⌚ 7 min ☆ 5.0 Campus Bites

RM 8.50

Add

Rice Bowls

ACTIVE

Teriyaki Chicken Rice Bowl

Grilled chicken, rice, vegetables,
sesame, and teriyaki glaze.

⌚ 12 min ☆ 5.0 Campus Bites

RM 12.90

Add

Drinks

ACTIVE

Milo Dinosaur

Iced Milo with extra malt powder,
a classic campus drink.

⌚ 3 min ☆ 0.0 Nasi Corner

RM 4.80

Add

Local Favorites

ACTIVE

Nasi Lemak Ayam

Coconut rice, sambal, egg,
cucumber, peanuts, and spiced
fried chicken.

⌚ 10 min ☆ 0.0 Nasi Corner

RM 11.90

Add

Cart and checkout

Campus Bites

MenuCartTrack

1 Hassan Basheir

CART

Checkout and schedule

Crispy Chicken Burger

Campus Bites

RM 13.50

-

1

+

Payment

Schedule pickup

As soon as possible

Payment method

TnG eWallet

Subtotal

RM 13.50

Total

RM 13.50

Place order

Order tracking

Campus Bites

MenuCartTrack

1 Hassan Basheir

ORDER TRACKING

Track queue status

Refresh

Campus Bites

RM 6.80

Cancelled

Nasi Corner

RM 11.90

Confirmed

Campus Bites

RM 17.00

Preparing

Campus Bites

RM 13.50

Ready

Campus Bites

RM 12.90

Completed

Order #E6F52B4B

Campus Bites

Ready

Confirmed

Preparing

Ready

Called

Completed

Queue number

A102

Estimated wait

24 min

1x Crispy Chicken Burger

RM 13.50

SEED-E6F52B4B

Paid

Vendor dashboard

 Campus Ops

Rashad Sofan

VENDOR

 Vendor Dashboard

Log out

VENDOR MANAGEMENT

Orders, inventory, and sales

Total orders

4

all seeded and live orders

Revenue

RM 43.40

paid mock transactions

Rating

5.00

average completed reviews

Live order queue

2 active

Hassan Basheir

A103

2x Campus Veggie Wrap

RM 17.00

Ready

Hassan Basheir

A102

1x Crispy Chicken Burger

RM 13.50

Ready

+ Add menu item

Name

Description

Category

Price

0

Prep minutes

8

Stock

10

Image URL

☒ Available to students

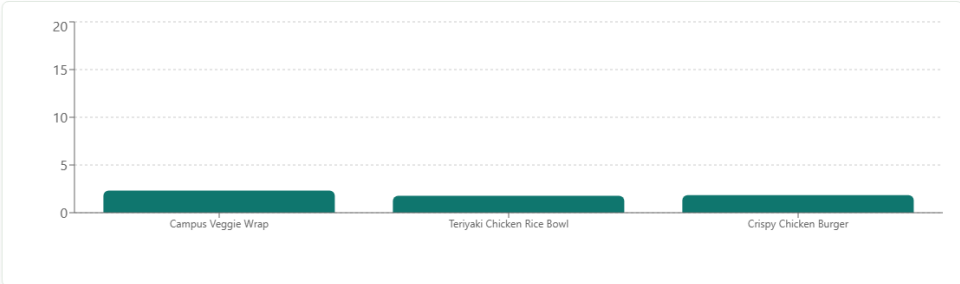
Save menu item

Menu and inventory

5 items

ITEM	PRICE	STOCK	STATUS	ACTIONS
Crispy Chicken Burger Burgers	RM 13.50	18	ACTIVE	Edit
Sold Out Brownie Box Dessert	RM 5.50	0	SUSPENDED	Edit
Iced Matcha Latte Drinks	RM 6.80	40	ACTIVE	Edit
Campus Veggie Wrap Grab & Go	RM 8.50	24	ACTIVE	Edit
Teriyaki Chicken Rice Bowl Rice Bowls	RM 12.90	35	ACTIVE	Edit

Sales analytics



Admin dashboard

Campus Ops

Ahmad Aljammal
ADMIN

Dashboard

Users

Reports

Settings

Log out

ADMINISTRATION

Campus dining control center

Export CSV

Users

5

all registered roles

Vendors

2

5 orders tracked

Revenue

RM 55.30

1 failed payments

Create vendor

Outlet name

Contact name

Email

Phone

Operating hours

Location

Initial password

08:00-20:00

Campus Food Cou

.....

Create vendor account

Create staff

Staff name

Email

Phone

Assigned outlet

Initial password

Campus Food Cot

.....

Create staff account

Reset password

User

New password

Ahmad Aljammal - ADMIN

Update password

Revenue by category

20

15

10

5

0

Grab & Go

Burgers

Local Favorites

Orders by status

4

3

2

1

0

CANCELLED

COMPLETED

CONFIRMED

PREPARING

READY

User and vendor management

5 records

All roles

NAME	ROLE	PROFILE	STATUS	UPDATE
Ahmad Aljammal admin@campus.test	ADMIN	Campus operations	ACTIVE	Active
Mohammed Alakklouk staff@campus.test	STAFF	Central Food Court	ACTIVE	Active
Hassan Basheir student@campus.test	STUDENT	251UC2506T	ACTIVE	Active
Nasi Corner Operator vendor2@campus.test	VENDOR	Nasi Corner	ACTIVE	Active
Rashad Sofan vendor@campus.test	VENDOR	Campus Bites	ACTIVE	Active

System configuration

mockPaymentSuccessRate

0.9

Probability that the mock payment gateway approves a payment.

operatingHours

08:00-22:00

Campus-wide ordering operating hours.

queuePrefix

A

Prefix used when queue tickets are generated.

reviewRequiredForRating

false

Whether a text review is required with a star rating.

serviceFeePercent

3

Campus service fee percentage shown in admin configuration.

Save settings

Staff queue panel

STAFF QUEUE

Call and confirm pickups

 Refresh

NOW SERVING



No ticket currently called

 Call next ready order

A102

Hassan Basheir

1x Crispy Chicken Burger

RM 13.50

Ready

Call

A103

Hassan Basheir

2x Campus Veggie Wrap

RM 17.00

Preparing

A104

Hassan Basheir

1x Nasi Lemak Ayam

RM 11.90

Confirmed

5 Implementation, Testing, User Guide, and Conclusion

The implementation uses React, Vite, TypeScript, Node.js, Express, SQLite, Socket.IO, Recharts, Zod, bcryptjs, JWT, Mermaid, Playwright, python-docx, and ReportLab.

Setup: npm install, npm run seed, npm run dev. Verification: npm run build, npm run test, npm audit --audit-level=high.

Demo accounts are student@campus.test, vendor@campus.test, admin@campus.test, and staff@campus.test with passwords documented in the DOCX and README. Vendor and staff accounts are created by the admin dashboard, not public registration.

The public queue display reads real queue tickets from the backend and updates when vendors mark orders ready or staff call a queue number.

The prototype is complete for local demonstration. Future work should integrate campus SSO, a sandbox payment gateway, production HTTPS deployment, and automated end-to-end tests.